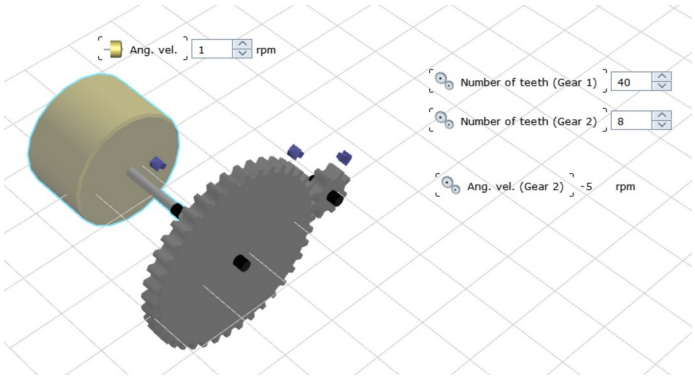
Simple Gear Train Worksheet

Q1.
You know how to build a simple gear train that will increase the speed of rotation of the driven gear compared to the driver gear.
From a selection of four gear wheels: 8 t, 16 t, 24 t and 40 t, design and build a simple gear train on Yenka that will provide the biggest increase in speed between the driver and driven gears.



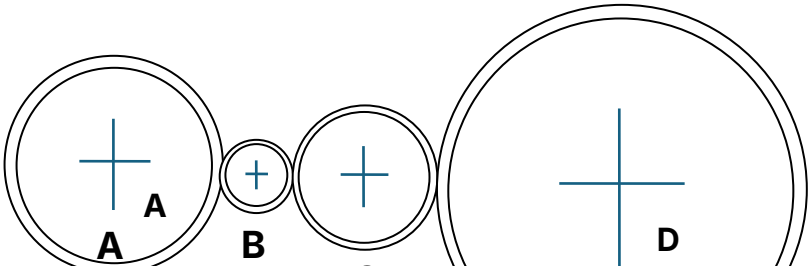
An increase in speed means the driven gear needs to be smaller than the driver.
To maximise the Velocity Ratio, VR, use the biggest gear available (40T) and smallest (8T) for the driver and driven gears respectively.

Calculate the velocity ratio of your system in your jotter.

Q2.
Modify your simple gear train so that it will give you the biggest decrease in speed between the driver and driven gears.
A decrease in speed means the driven gear needs to be larger than the driver.
To maximise the Velocity Ratio, VR, use the biggest gear available (40T) and smallest (8T) for the driven and driver gears respectively.

Calculate the velocity ratio of your system in your jotter.

Q3. In your jotter for this simple gear train, calculate the following:



A = 50 teeth
B = 10 teeth
C = 25 teeth
D = 100 teeth

a) Which gear rotates in the same direction as A?
Gear C

b) The velocity ratio of A to B, A to C and at D

i. A to B

ii. A to C

iii. A to D

c) The speed of gear B, C and D if A rotates at 500 revs min^{-1}.

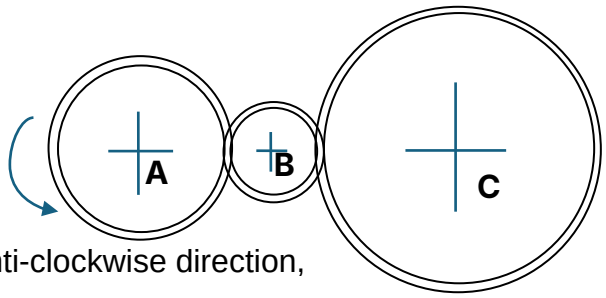
i. B =

ii. C=

iii. D=

Q4. A simple gear train is shown.

A = 20 teeth
B = 5 teeth
C = 30 teeth

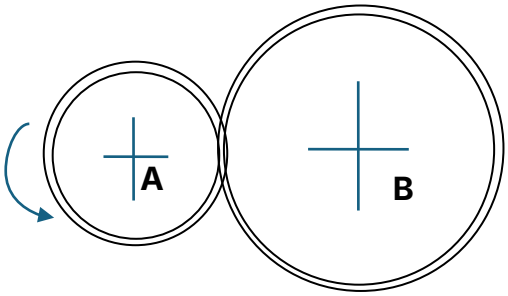


If gear A rotates at 250 revs min^{-1} in a anti-clockwise direction, calculate;

(a) The Velocity Ratio of the gear system.

(b) The output speed.

Q5. A simple gear train is shown. The driver gear A has 20 teeth. When shaft A is rotated 10 times, shaft B rotates five times.



(a) How many teeth has gear B?

40 teeth
(If B rotates ½ the number of turns of A, it is going slower and therefore must be larger – have more teeth than A)

(b) What is the velocity ratio of the system?

Because we are working out a ratio, it does not matter if they do these turns in a second or in a day. But let's assume that the 10 and 5 rotations by gears A and B happen in a minute to keep things simple.

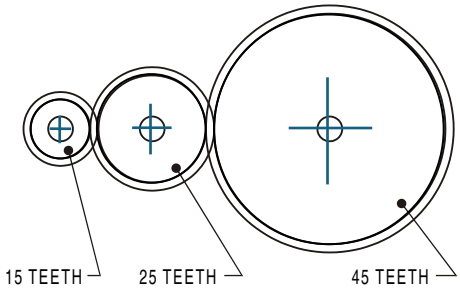
Or, we could use the number of teeth worked out in part a.

(c) If shaft A rotates at 60 revs min^{-1}, at what speed does shaft B rotate?

(d) If shaft A rotates anti-clockwise, in which direction does shaft B rotate?
Clockwise

Q6. The gear with 15 teeth is driven by a motor at 1800revs min^{-1}. The gear with 25 teeth is an idler gear.

(a) What is the velocity ratio of the gear system shown?



(b) If the driver rotates at 1800 revs min^{-1}, at what speed does the largest gear rotate?

(c) If the driver rotates anti-clockwise, in which direction does the output rotate?
Anti-clockwise also.